A Zero-Parameter Framework for Cosmological Dynamics and Galaxy Cluster Mass Discrepancies via Structural Self-Energy Expansion (SSEE-V3.6)

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Abstract

We present the Structural Self-Energy Expansion (SSEE-V3.6), a minimalist theoretical framework that unifies cosmological expansion and cluster dynamics using dimensionless geometric constants (π and the golden ratio Φ) without introducing free parameters or exotic dark-matter particles. The framework defines a closed algebraic system yielding an asymptotic structural viscosity factor $\mathrm{KAL}_0 \approx 5.5214$, natively reproducing the observed cosmic dark-to-baryonic matter density ratio. At galactic scales, SSEE derives an acceleration threshold $a_0 = c\,H_0/\mathrm{KAL}_0 \approx 1.186 \times 10^{-10}\,\mathrm{m\,s^{-2}}$. On cosmological scales, the model predicts a dynamical dark-energy equation of state with $w_0 \approx -0.840$ and $w_a \approx -0.670$, providing a parsimonious alignment with dynamical trends in the DESI DR2 observations [Abdul-Karim et al., 2025]. Incorporating an empirical light-neutrino mass sum $\sum m_\nu = 0.085\,\mathrm{eV}$, the framework structurally resolves the megaparsec cluster mass discrepancy when combined with a variable Integrated Galaxy-wide Initial Mass Function (IGIMF) [Zhang et al., 2026]. A full Bayesian MCMC validation against DESI DR2, Planck 2018, and galaxy-cluster mass data is presented in the companion Paper 2 [Almeida Vallejo, 2026].

1 Introduction and Theoretical Foundations

The SSEE framework operates under the principle of algebraic closure and macroscopic structural self-energy. Rather than adding particle degrees of freedom, the system's identity is defined by a closed set of geometric invariants derived exclusively from π and Φ. These fundamental scales dictate the macroscopic behaviour of spacetime:

- $\boldsymbol{\Omega}$ ($\pi + \Phi \approx 4.7596$): Stability Metric.
- $\boldsymbol{\beta}$ ($(\pi + \Phi)/2 \approx 2.3798$): Base Coupling Scalar.

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- $\mathbf{KAL_0}$ ($\beta + \pi \approx 5.5214$): Asymptotic Structural Viscosity.
- $\boldsymbol{P}$ ($\Omega + \Phi \approx 6.3776$): Dynamical Evolution Scalar.
- $\boldsymbol{K_v}$ ($\Phi + \pi + \Omega \approx 9.5192$): Structural Constraint Invariant.
- $\boldsymbol{T_r}$ ($3(\Phi + \beta) \approx 11.9935$): 3D Saturation Horizon.
- $\boldsymbol{M_v}$ ($\Phi + \pi + K_v \approx 14.2788$): Maximal Dimensional Invariant.

1.1 Axiomatic Principles

SSEE is explicitly constructed as a top-down theoretical ansatz. Its internal symmetries are not phenomenological fits derived bottom-up, but fundamental axiomatic identities that the macroscopic universe must obey. Three physical postulates govern the interactions of these geometric invariants:

1. **The Temporal Triad (Cosmological Epochs):** Spacetime evolves through three distinct thermodynamic states — the Origin/Law phase (initial singularity), the Ignition/Processing phase (structure formation), and the Transcendental Saturation phase (dark energy asymptotic domination). The Hubble history $H(a)$ is bounded by:
$$\lim_{a\to 0} H^2(a) \propto \rho_{\rm bar}, \qquad \lim_{a\to\infty} H^2(a) = \frac{8\pi G}{3}\,\rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right). \tag{1}$$
2. **Non-Self-Addition (Evolution by Difference):** Structural evolution proceeds through non-linear geometric scaling rather than linear superposition. All macroscopic observables $\mathcal{O}$ must be strictly functional dependencies of the invariant set and H_0; no additional free parameters are permitted:
$$\mathcal{O} = f(\Omega,\, \beta,\, \mathrm{KAL}_0,\, P,\, K_v,\, T_r,\, M_v;\, H_0). \tag{2}$$
3. **Asymmetric Equivalence:** Energy states can possess identical geometric values but divergent dynamic functions. The macroscopic geometric response must inversely mirror the physical source:
$$\mu_{\rm SSEE}(x) \cdot \mathrm{KAL}(x) \equiv 1. \tag{3}$$

1.2 The ssee Effective Action

Instead of an arbitrary cosmological constant Λ, the vacuum is governed by a static geometric pressure field tied to the critical density $\rho_{\rm crit,0} = 3H_0^2/(8\pi G)$:
$$S_{\rm SSEE} = \int d^4x\,\sqrt{-g}\left[\frac{R}{16\pi G} - \rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right) + \mathcal{L}_{\rm matter}\right]. \tag{4}$$
This action guarantees that the geometric constraints are not phenomenological fits, but inevitable consequences of the universe's macroscopic structural self-energy at saturation.

Falsifiability. The framework is strictly falsifiable. It would be refuted by: (1) direct detection of a collisionless dark-matter particle accounting for the missing mass; (2) definitive cosmological constraints proving $w_a = 0$; or (3) high-significance confirmation of a normal neutrino hierarchy with $\sum m_\nu < 0.06\,\mathrm{eV}$.

2 Dark Energy Evolution

SSEE models dynamical dark energy via the Chevallier–Polarski–Linder (CPL) form $w(a) = w_0 + w_a(1-a)$. Motivated by Eq. (4), the expansion is subject to an effective geometric pressure:

$$H^2 = \frac{8\pi G}{3}(\rho_{\rm bar} + \rho_{\rm DE}), \qquad \rho_{\rm DE} = \rho_{\rm crit,0}\left(\frac{T_r}{M_v}\right). \tag{5}$$

The baseline state w_0 is set by the static geometric ratio T_r/M_v, and the evolution w_a encodes the dynamical departure from saturation driven by P/K_v:

$$w_0 = -\frac{T_r}{M_v} \approx -0.840, \qquad w_a = -\frac{P}{K_v} \approx -0.670. \tag{6}$$

Asymptotic saturation. Although $w_0 + w_a = -1.510$ would ordinarily imply a phantom Big Rip, SSEE adopts CPL only as a local observational parameterisation. In the far future ($a \to \infty$) the energy density saturates at the finite T_r/M_v boundary of Eq. (4), ensuring universal stability.

DESI DR2 [Abdul-Karim et al., 2025] analyses indicate an evolving preference for dynamical dark energy. The SSEE algebraic point $(w_0, w_a) = (-0.840, -0.670)$ falls within the 68% confidence contours of the combined BAO+CMB+DESY5 datasets (Mahalanobis $\chi^2_{2D} = 0.080$, i.e. 0.05σ; quantified in Paper 2 [Almeida Vallejo, 2026]).

3 Galaxy Cluster Dynamics

3.1 The Cluster Mass Discrepancy and IGIMF

In rich galaxy clusters, inferred dynamical mass typically exceeds the visible baryonic mass by a factor of ∼5–6. Recent literature incorporating a variable IGIMF [Zhang et al., 2026] demonstrates that baryonic masses have been systematically underestimated: heavy stellar remnants raise the effective baryonic baseline to at least $88^{+5}_{-4}\%$ of the MOND dynamical mass. In SSEE, this updated IGIMF baseline multiplied by $\mathrm{KAL}_0 \approx 5.5214$ natively closes the remaining discrepancy without cold dark matter.

3.2 Structural Viscosity and the Neutrino Bridge

At the cluster scale, the total effective dynamical mass $M_{\rm dyn}$ within radius R is:

$$M_{\rm dyn}(R) = M_{\rm bar}^{\rm IGIMF}(R) \cdot \mathrm{KAL}_0 \cdot [1 + f_\nu], \tag{7}$$

where f_ν quantifies the fractional mass contribution of trapped relic neutrinos. Using $\sum m_\nu = 0.085\,\mathrm{eV}$ — robustly permitted within SSEE's dynamic dark-energy regime — the cosmological neutrino density is $\rho_\nu \approx 28.6\,\mathrm{eV\,cm^{-3}}$. For a cluster with escape velocity $v_{\rm esc} \approx 2000\,\mathrm{km\,s^{-1}}$, integration of the Fermi–Dirac distribution [Tremaine & Gunn, 1979] yields a trapped fraction $f_\nu \approx 0.018$–0.022. This structurally consistent ∼2% correction provides the final macroscopic bridge.

3.3 Quantitative Cluster Analysis

Applying Eq. (7) to four well-documented clusters demonstrates predictive accuracy against observed dynamical masses (Table 1).

Table 1: Cluster mass predictions. All masses within R_{500} in units of $10^{14}\,M_\odot$. $M_{\rm SSEE} = M_{\rm bar}^{\rm IGIMF} \times 5.5214 \times 1.02$. IGIMF baselines from Zhang et al. [2026]; observed masses from Simionescu et al. 2011, Walker et al. 2012, Pointecouteau et al. 2004, Clowe et al. 2006.

Cluster	$M_{\rm bar}^{\rm IGIMF}$ $[10^{14}\,M_\odot]$	$M_{\rm SSEE}$ $[10^{14}\,M_\odot]$	$M_{\rm dyn}^{\rm obs}$ $[10^{14}\,M_\odot]$
Coma	1.8 ± 0.2	10.1 ± 1.1	9.8 ± 1.0
A2029	2.2 ± 0.2	12.4 ± 1.1	12.0 ± 1.2
A478	1.5 ± 0.2	8.4 ± 1.1	8.0 ± 1.0
Bullet (main)	1.2 ± 0.2	6.7 ± 1.1	6.5 ± 1.0

All four predictions agree with observed masses at the 1σ level ($\chi^2_r = 0.122$; full analysis in Paper 2 [Almeida Vallejo, 2026]).

3.4 Gravitational Lensing and the Newtonian Limit

The Bullet Cluster (1E 0657-56) is frequently cited as proof of particulate dark matter due to the spatial offset between collisional X-ray gas and lensing-mass peaks [Clowe et al., 2006]. SSEE reproduces this topological offset through macroscopic structural viscosity without collisionless particles.

The structural viscosity acts as an environment-dependent scalar field. Defining $x = |\nabla\Phi|/a_0$, SSEE adopts an interpolation function analogous to the canonical MOND simple interpolating function $\mu(x)$:

$$\mu_{\rm SSEE}(x) = \frac{x+1}{x+{\rm KAL}_0} \implies {\rm KAL}(x) = \frac{1}{\mu_{\rm SSEE}(x)} = \frac{x+{\rm KAL}_0}{x+1}. \tag{8}$$

For $x \gg 1$ (Newtonian limit), ${\rm KAL}(x) \to 1$. For $x \ll 1$ (deep MONDian limit), ${\rm KAL}(x) \to {\rm KAL}_0$. During a high-velocity cluster collision, the stripped X-ray gas flattens its local potential gradient ($x \ll 1$), activating the full ${\rm KAL}_0$ amplification; collisionless galaxies retain compact potential wells ($x \geq 1$), coupling non-linearly.

The effective lensing convergence is:

$$\kappa(\vec{\theta}) = \frac{\Sigma_{\rm SSEE}(\vec{\theta})}{\Sigma_{\rm crit}}, \quad \Sigma_{\rm SSEE}(\vec{\theta}) = \int \rho_{\rm bar}^{\rm IGIMF}(\vec{\theta},\ell)\cdot {\rm KAL}(x)\,d\ell, \quad \Sigma_{\rm crit} = \frac{c^2}{4\pi G}\frac{D_S}{D_L D_{LS}}. \tag{9}$$

This allows lensing peaks to track galaxies, consistent with QUMOND simulations [Angus et al., 2007].

Data Availability

All observational data used in this work are publicly available. DESI DR2 BAO measurements are from Abdul-Karim et al. [2025] (`https://arxiv.org/abs/2503.14738`). Galaxy-cluster dynamical masses are from Zhang et al. [2026] (`https://arxiv.org/abs/2602.06082`). Analysis scripts and data files are available at `https://github.com/mikealmeida1721/SSEE`.

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